Data Engineering Project - Milestone M0: Data Source Proposal

Course: Introduction to Data Science and Engineering | **Submission:** 07.10.2026

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Team note: I do not have teammates yet. I have contacted the instructor about team formation and have not received a reply so far. I am submitting on time and will add teammates' names as soon as the team is confirmed.

Candidate 1 (preferred): Deezer API + MusicBrainz

1.1 Source identification

- **Sources:** Deezer public API (docs: https://developers.deezer.com/api, base URL `https://api.deezer.com`) and the MusicBrainz web service (https://musicbrainz.org/doc/MusicBrainz_API).
- **Access:** Public REST APIs returning JSON. Deezer chart/artist/album endpoints need no authentication. MusicBrainz needs no key but requires a User-Agent header with contact info and allows about 1 request per second.
- **Access verified:** Yes. I called `https://api.deezer.com/chart/0/tracks?limit=100` and received 100 track records.

1.2 Data description

- **One row =** (a) a track at a given chart position on a given snapshot date; (b) an artist with its fan count; (c) an album with release date and genre; (d) a MusicBrainz recording/artist with metadata (country, tags, first release date).
- **Records:** Each chart call returns up to 100 tracks, and charts exist per genre, so one snapshot across genre charts is expected to exceed 1,000 chart rows, plus the artist and album records they reference. The global chart alone adds 100 rows per daily snapshot.
- **Fields (observed):** track `id` (int), `title`, `title_short` (string), `rank` (int), `position` (int), nested `artist` (`id`, `name`) and `album` (`id`, `title`). **Expected from artist/album endpoints:** `nb_fan` (int), `release_date` (date), `nb_tracks` (int), `genres`.
- **Update frequency:** Charts change daily; I will take one snapshot per day with a scheduled script.
- **Joins:** `chart.artist.id = artist.id`, `chart.album.id = album.id`; Deezer tracks to MusicBrainz recordings via ISRC (from the track endpoint) with name matching as a fallback.

1.3 Project intent

Can we predict whether a track that enters a Deezer genre chart will still be in the top 20 (or climb) seven days later, using artist fan count, album age, genre, track duration and explicit-content flag? Beyond prediction, the project would explain which factors (established artist popularity vs. release recency vs. genre) drive chart longevity. A label, playlist curator or artist manager could use this to decide which new releases deserve promotion. MusicBrainz metadata (artist country, tags) adds context that Deezer alone does not provide.

1.4 Anticipated difficulties

The chart endpoint only returns the current ranking, not history, so the history must be built by collecting daily snapshots starting immediately. Deezer (about 50 requests per 5 seconds) and MusicBrainz (1 request per second) both rate-limit, so enrichment calls must be throttled and cached. Matching tracks to MusicBrainz by name is noisy, and some nested fields may be missing or null.

Candidate 2: ListenBrainz + MusicBrainz

2.1 Source identification

- **Sources:** ListenBrainz API (docs: https://listenbrainz.readthedocs.io/en/latest/users/api/index.html, root `https://api.listenbrainz.org`) and MusicBrainz (as above).
- **Access:** Public REST API, rate-limited via response headers. A free account user token can be sent in the Authorization header for higher rate limits. **Status:** token not yet obtained and endpoints not yet tested, which is why this is ranked second.

2.2 Data description

- **One row =** a recording or artist with its aggregate listen count for a time range (sitewide statistics and popularity endpoints), or a MusicBrainz recording/artist with metadata. Only aggregate data is used, never individual users' listening histories.
- **Records:** Statistics and popularity endpoints cover thousands of distinct artists and recordings; public data dumps also exist. Exact counts will be confirmed once endpoints are tested.
- **Fields (expected):** MBIDs (UUID), names, listen counts, listener counts, time range; MusicBrainz adds country, release dates and tags.
- **Update frequency:** Listens arrive continuously; statistics are recomputed on a regular schedule.
- **Joins:** MusicBrainz ID (MBID) between ListenBrainz statistics and MusicBrainz metadata.

2.3 Project intent

Which recordings will gain the most listens next month, based on this month's listen growth, release year, artist country and tags? The project would also explain how much metadata (age, origin, genre tags) matters compared to past listening momentum. Streaming services and independent artists could use it to spot emerging tracks early.

2.4 Anticipated difficulties

Rate limits and token handling need care, and some listens are not mapped to MBIDs, which creates coverage gaps. MusicBrainz's 1 request per second limit makes bulk metadata enrichment slow. The exact time-range semantics of the statistics endpoints need to be understood before they are used as time series.